

N8n Home assignment

Home Assignment 2:

Build a Document Question Answering (Document QA) Agent

Develop an AI-powered Document QA agent that enables users to upload a document from their local system and ask questions based on its content. The agent should extract the document content, understand the user's query, and provide accurate answers using only the uploaded document as the source of information. If the requested information is not available in the document, the agent should respond appropriately instead of generating unsupported answers.

Expected Outcome: Users should be able to upload a document, ask multiple questions, and receive context-aware answers derived only from the uploaded document.

list of nodes :

1.Manua Trigger

2.Read /write files from disk

3.Extract from file

4.Basic LLM chain

Pre-requite :

Create a folder 'n8n-files' in the below path where n8n is downloaded(already .n8n folder is present in the below path as it is downloaded)

C:\Users\rta

Place the file inside the new folder you created

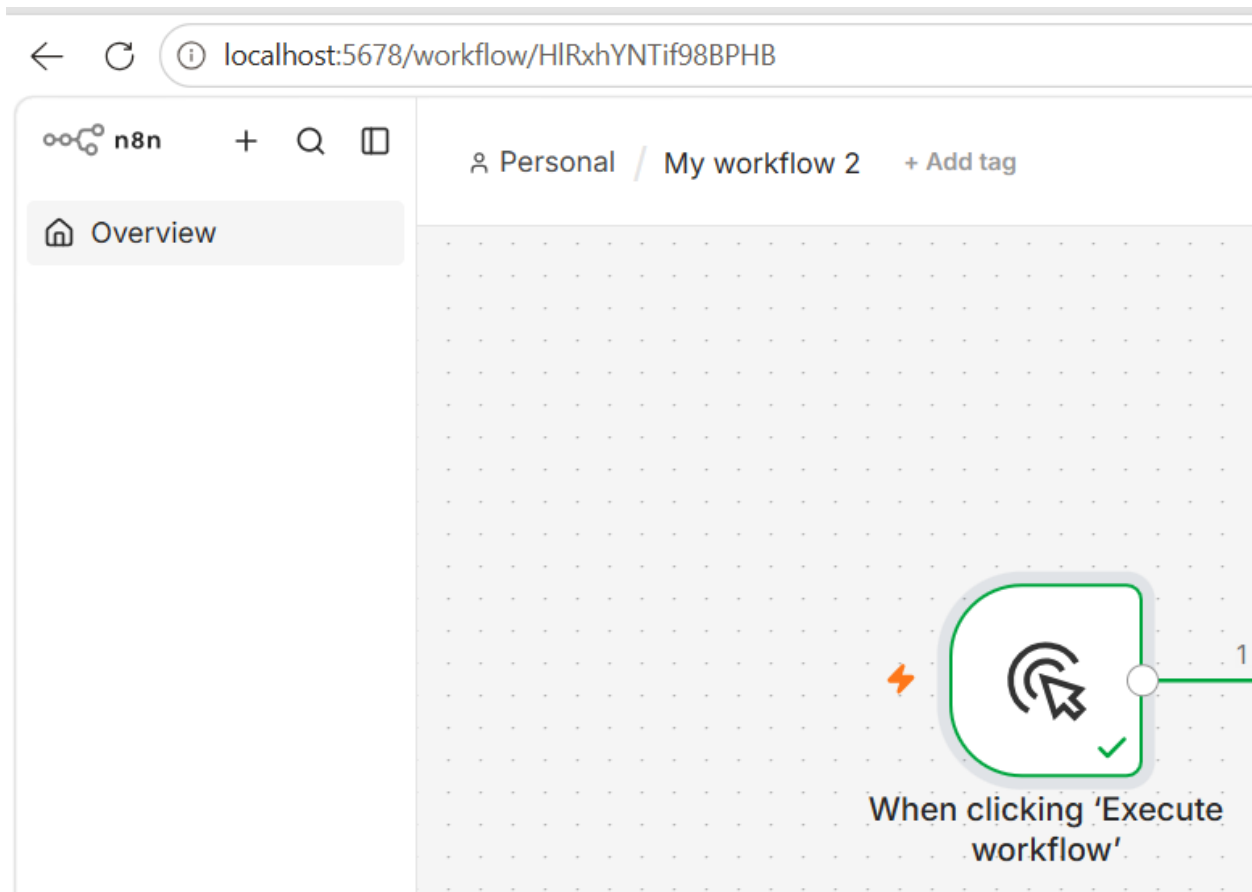
C:\Users\rta\n8n-files

If you don't want to share the file to n8n as it is confidential, follow the below steps

1. In windows search for 'Edit the system environment variable' and click on open.
2. Click on 'Environment variable button'
3. Click on 'New' under 'user variables for rta'
4. Give the path where u had saved the document
C:\Users\rta\Pictures

Node 1 : Manual Trigger

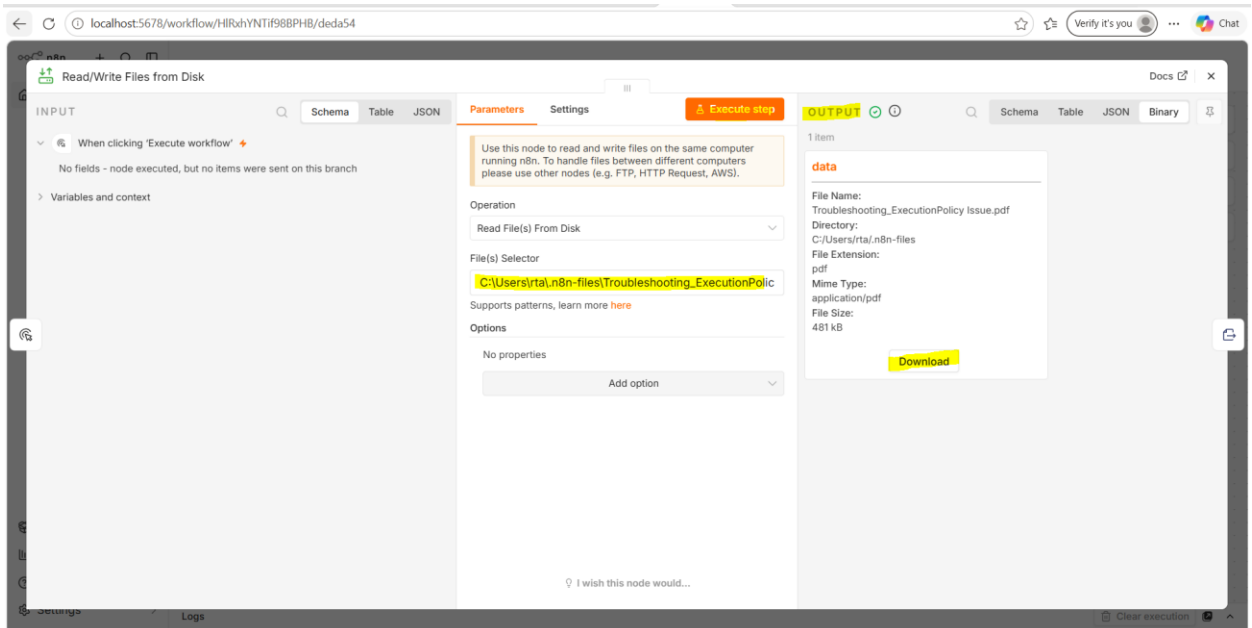
1. Click on '+' icon on the right and search for 'Manual Trigger'
2. Select it and the node will be added successfully



Node 2 : Read /write files from disk

1. Click on '+' icon on the right and search for 'Read /write files from disk'
2. Select 'Read from disk' and the node will be added successfully
3. Double click on this node and give the path of the file where you have placed along with the name of the file.pdf as it is a pdf file.
C:\Users\rta\.n8n-files\Troubleshooting_ExecutionPolicy Issue.pdf

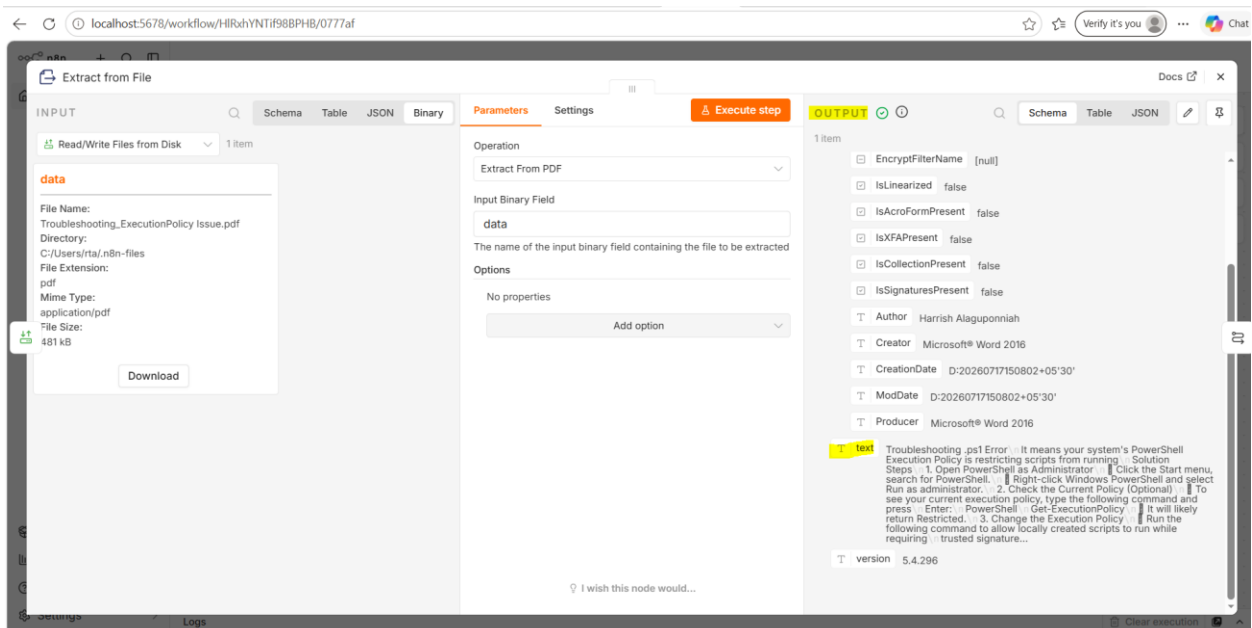
4. Click on 'Execute Step'



5. Close the node

Node 3 : Extract from file

1. Click on '+' icon on the right and search for 'Extract from file'
2. Select it and the node will be added successfully
3. Double click on the node
4. Click on 'Execute Step'
5. Verify if the output has the 'Text' section to make sure the text from the given pdf document is read as shown in the below screenshot



6. Close the node

Node 4 : Basic LLM chain

1. Click on '+' icon on the right and search for 'Basic LLM Chain'
2. Select it and the node will be added successfully
3. Double click on the node
4. Select 'Define Below' from the drop-down
5. Click on 'Expression' and give the below prompt

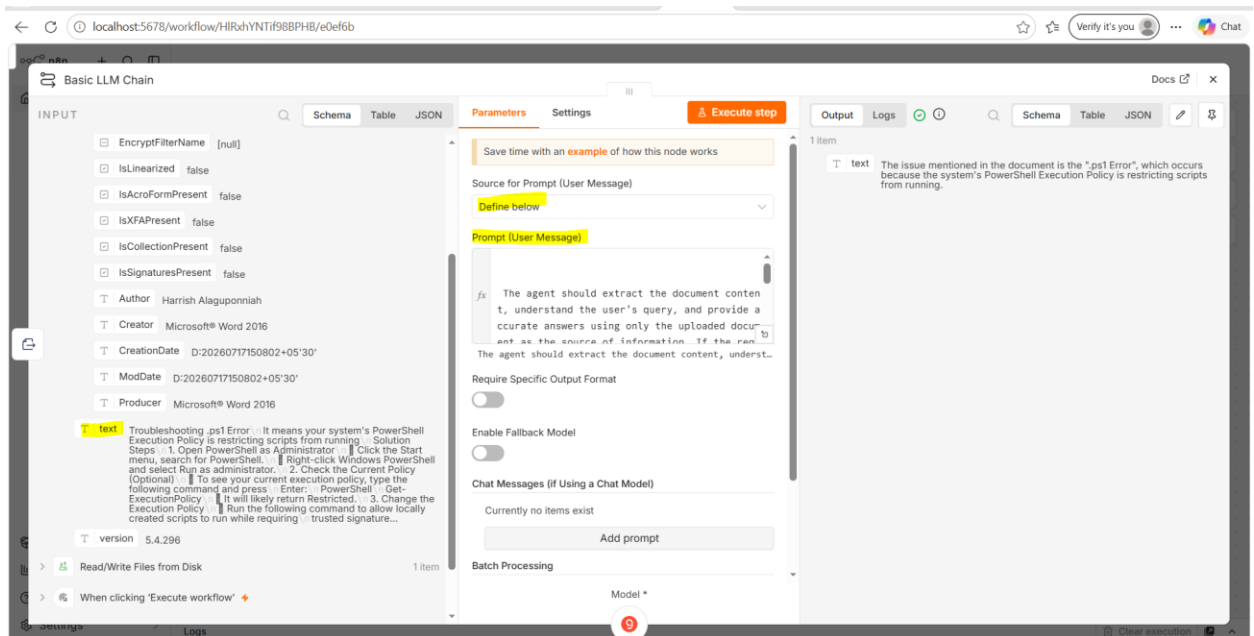
The agent should extract the document content, understand the user's query, and provide accurate answers using only the uploaded document as the source of information. If the requested information is not available in the document, the agent should respond appropriately instead of generating unsupported answers.

[Strict] Don't generate responses based on the general knowledge. Dont hallucinate the details. Just answer only based on the document that is provided. If the question asked is not on the document, provide a message 'question asked is not relevant to the document'

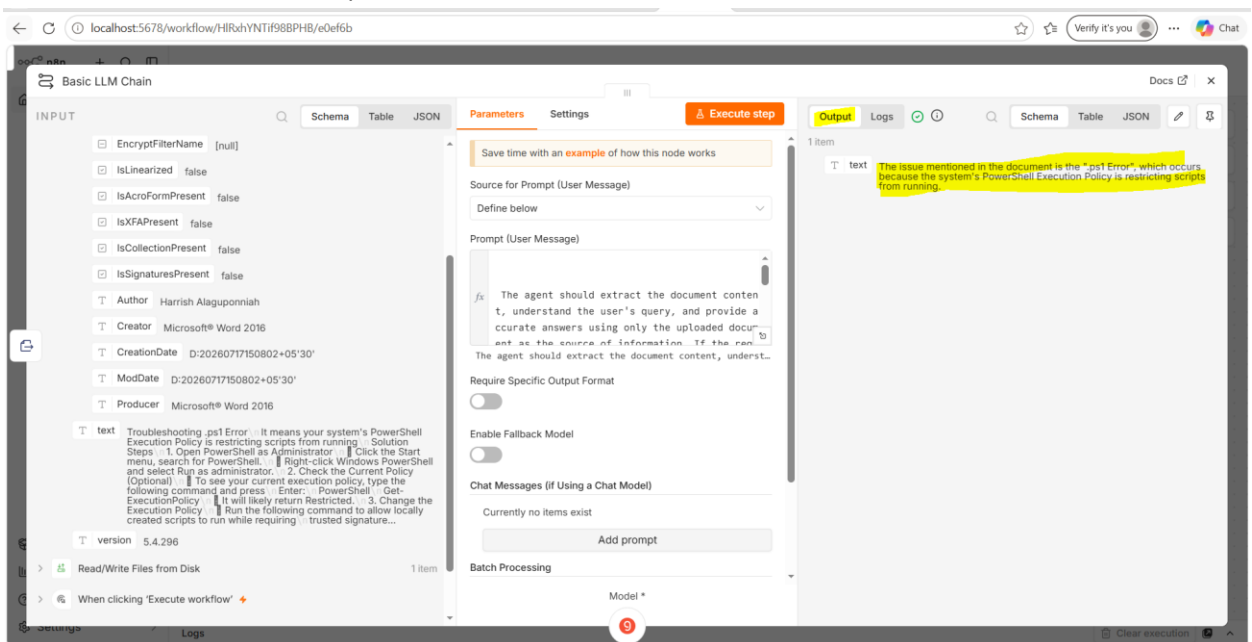
Document Content : {{ \$json.text }}

Input :

What is the issue mentioned in the document



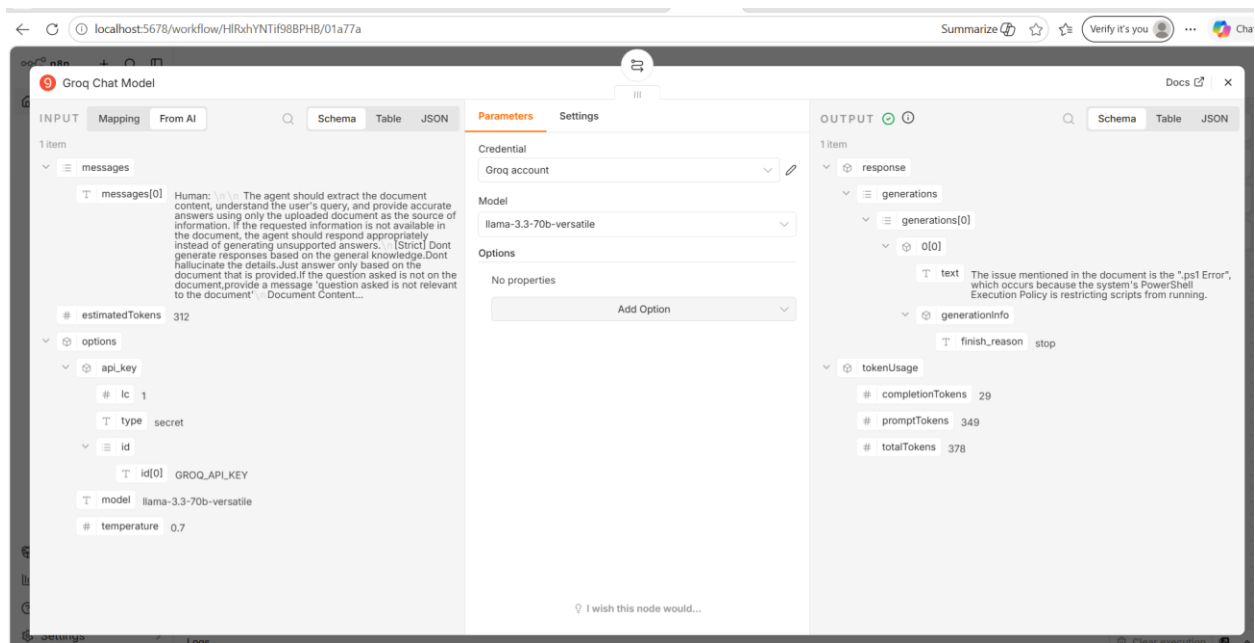
6. Document Content : `{{ $json.text }}` (In the prompt)
Drag and drop the highlighted one from the left side where there is 'Text' as shown in the above screenshot
7. Click on the delete icon on the left side of User field
8. Click on 'Execute Step'
9. Relevant answers from the PDF should be displayed in the output
10. Question in the prompt is 'What is the issue mentioned in the document(given in the prompt).'
11. Answer is received in the output



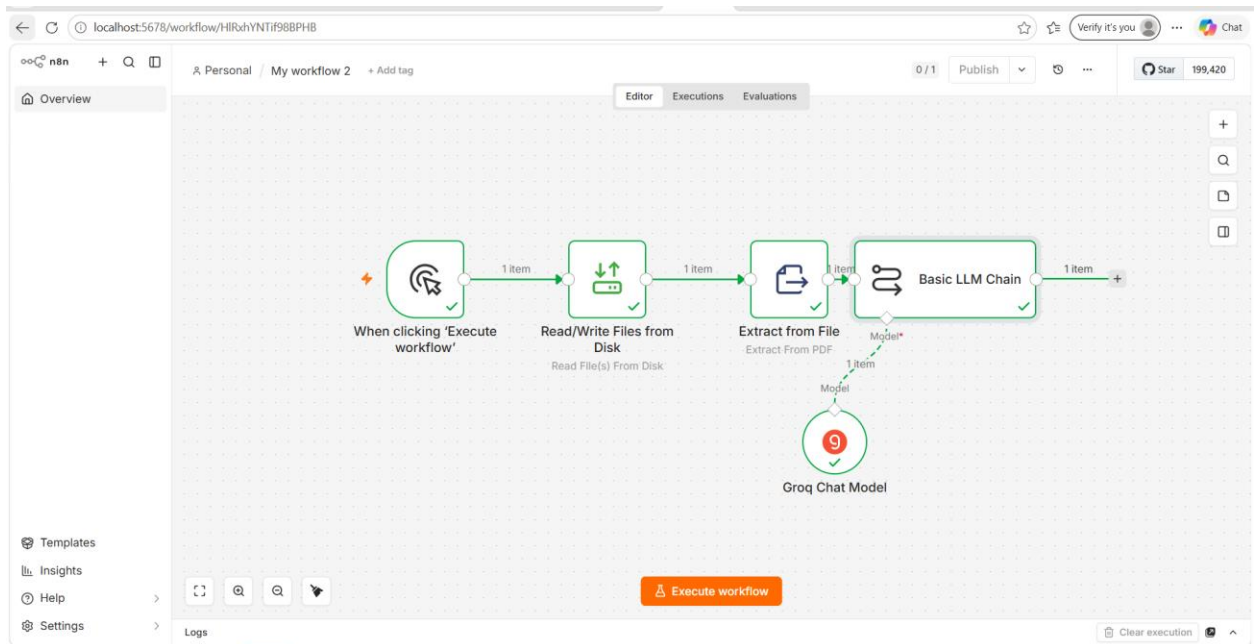
The image displays two screenshots. The top screenshot shows a workflow editor interface for a 'Basic LLM Chain' node. The 'Parameters' tab is active, showing input fields for 'numpages' (2), 'numrender' (2), and 'info'. The 'Prompt (User Message)' field contains a detailed instruction for an agent to extract document content and provide accurate answers. The 'Output' section shows a sample response from the LLM regarding a PowerShell error. The bottom screenshot shows a document viewer interface with a sidebar of tools. The main content area displays a document titled 'Troubleshooting .ps1 Error' with a detailed solution for a PowerShell execution policy issue, including steps to open PowerShell as Administrator and check the current policy.

Node 5 : Groq chat model

1. Click on '+' icon on the right and search for 'Groq'
2. Select it and the node will be added successfully
3. Double click on this node
4. Create a new API key in 'Console.groq' website. Copy it and paste it in this node.
5. Select the model as 'llama-3.3-70b-versatile'
6. Close this node



Overall Execution flow



Additional Questions :

What is the first solution step given in the document to resolve the issue?

localhost:5678/workflow/HIRxhYNTiF98BPHB/e0ef6b

Basic LLM Chain

INPUT

Extract from File

numpages 2

numrender 2

Info

PDFFormatVersion 1.5

Language en-US

EncryptFilterName [null]

IsLinearized false

IsAcroFormPresent false

IsXFAPresent false

IsCollectionPresent false

IsSignaturesPresent false

Author Harrish Alaguponniah

Creator Microsoft® Word 2016

CreationDate D:20260717150802+05'30'

ModDate D:20260717150802+05'30'

Producer Microsoft® Word 2016

Text Troubleshooting.ps1 Error: It means your system's PowerShell Execution Policy is restricting scripts from running. Solution Steps: 1. Open PowerShell as Administrator. 2. Click the Start menu, search for PowerShell, right-click Windows PowerShell, and selecting "Run as administrator".

Parameters

Save time with an example of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

In the document, provide a message 'question asked is not relevant to the document'.

Document Content : {{ \$json.text }}

Input :

What is the first solution step given in the document to resolve the issue?

The agent should extract the document content, understand the content, and provide the first solution step.

Require Specific Output Format

Enable Fallback Model

Chat Messages (if Using a Chat Model)

Currently no items exist

Add prompt

Batch Processing

Model *

Output

1 item

Text The first solution step given in the document to resolve the problem is to "Open PowerShell as Administrator". This involves clicking the Start menu, searching for PowerShell, right-clicking Windows PowerShell, and selecting "Run as administrator".

All tools Edit Convert E-Sign

Find text or tools

Share Ask AI Assistant

All tools

Export a PDF

Edit a PDF

Create a PDF

Combine files

Organize pages

AI Assistant

Generative summary

Request e-signatures

Scan & OCR

Protect a PDF

Redact a PDF

Compress a PDF

Prepare a form

View more

Convert, edit and e-sign PDF forms & agreements

Free trial

more information, see about Execution Policies at <https://go.microsoft.com/fwlink/?linkid=918830>.

At line:1 char:1

+ npm init playwright@latest

~

+ CategoryInfo : SecurityError: ([]) PSObject[]

+ FullExceptionMessage : UnauthorizedAccess

It means your system's PowerShell **Execution Policy** is restricting scripts from running

Solution Steps

- Open PowerShell as Administrator**
 - Click the **Start** menu, search for **PowerShell**.
 - Right-click **Windows PowerShell** and select **Run as administrator**.
- Check the Current Policy (Optional)**
 - To see your current execution policy, type the following command and press **Enter**:


```
PowerShell
Get-ExecutionPolicy
```

 - It will likely return **Restricted**.
- Change the Execution Policy**
 - Run the following command to allow locally created scripts to run while requiring trusted signatures for downloaded scripts:


```
PowerShell
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Which is the optional step mentioned in the document?

localhost:5678/workflow/HIRshYNTif98BPHB/e0ef6b

Basic LLM Chain

INPUT

Extract from File

numpages 2

numrender 2

Info

T PDFFormatVersion 1.5

T Language en-US

EncryptFilterName [null]

IsLinearized false

IsAcroFormPresent false

IsXFAPresent false

IsCollectionPresent false

IsSignaturesPresent false

T Author Harish Alaguponniah

T Creator Microsoft® Word 2016

T CreationDate D:20260717150802+05'30'

T ModDate D:20260717150802+05'30'

T Producer Microsoft® Word 2016

T text Troubleshooting .ps1 Error : It means your system's PowerShell Execution Policy is restricting scripts from running. Solution Steps : 1. Open PowerShell as Administrator : Click the Start menu, search for PowerShell. Right-click Windows PowerShell

Parameters

Save time with an example of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

In the document, provide a message 'question asked is not relevant to the document'

Document Content : ({{ \$json.text }})

Input :

Which is the optional step mentioned in the document?

The agent should extract the document content, understand...

Require Specific Output Format

Enable Fallback Model

Chat Messages (if Using a Chat Model)

Currently no items exist

Add prompt

Batch Processing

Model *

Output

1 item

T text The optional step mentioned in the document is to "Check the Current Policy". This is step 2, which involves typing the command "Get-ExecutionPolicy" in PowerShell to see the current execution policy.

All tools

Export a PDF

Edit a PDF

Create a PDF

Combine files

Organize pages

AI Assistant

Generative summary

Request e-signatures

Scan & OCR

Protect a PDF

Redact a PDF

Compress a PDF

Prepare a form

View more

Convert, edit and e-sign PDF forms & agreements

Free trial

Find text or tools

Share

Ask AI Assistant

PowerShell : File C:\Program Files\Nodejs\npm.ps1 cannot be loaded because running scripts is disabled on this system. For more information, see about_Execution_Policies at https://go.microsoft.com/fwlink/?LinkID=135170.

At line:1 char:1

+ npm init playwright@latest

+ ~~~~~

+ CategoryInfo : SecurityError: (1) [], PowerShellException

+ FullyQualifiedErrorId : UnauthorizedAccess

It means your system's PowerShell Execution Policy is restricting scripts from running

Solution Steps

1. **Open PowerShell as Administrator**
 - Click the Start menu, search for **PowerShell**.
 - Right-click **Windows PowerShell** and select **Run as administrator**.
2. **Check the Current Policy (Optional)**
 - To see your current execution policy, type the following command and press **Enter**:

PowerShell

Get-ExecutionPolicy

 - It will likely return **Restricted**.
3. **Change the Execution Policy**

Ask AI Assistant

Break this down into simple steps

By using AI Assistant, you agree to Generative AI User Guidelines

What information is provided in the document regarding AI?

localhost:5678/workflow/HIRxhYNTIf98BPHB/e0ef6b

Basic LLM Chain

INPUT

Extract from File

- # numpages 2
- # numrender 2

Info

- T PDFFormatVersion 1.5
- T Language en-US
- E EncryptFilterName [null]
- E IsLinearized false
- E IsAcroFormPresent false
- E IsXFAPresent false
- E IsCollectionPresent false
- E IsSignaturesPresent false
- T Author Harrish Alaguponniah
- T Creator Microsoft® Word 2016
- T CreationDate D:20260717150802+05'30'
- T ModDate D:20260717150802+05'30'
- T Producer Microsoft® Word 2016

Text

Troubleshooting.ps1 Error: It means your system's PowerShell Execution Policy is restricting scripts from running. Solution Steps: 1. Open PowerShell as Administrator. 2. Click the Start menu, search for PowerShell. 3. Right-click Windows PowerShell

Parameters

Save time with an example of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

In the document, provide a message 'question asked is not relevant to the document'

Document Content : `{{ $json.text }}`

Input :

What information is provided in the document regarding AI?

The agent should extract the document content, understand...

Require Specific Output Format

Enable Fallback Model

Chat Messages (if Using a Chat Model)

Currently no items exist

Add prompt

Batch Processing

Model *

Output

1 item

T text The question asked is not relevant to the document.